

| <b>RADIONUCLIDE SAFETY DATA SHEET</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                    |                                                                                                                                                                                                                                                                                             |         |   |
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| <b>RADIONUCLIDE: Mn-54</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                    | <b>FORMS: Soluble</b>                                                                                                                                                                                                                                                                       |         |   |
| <b>PHYSICAL CHARACTERISTICS</b><br>HALF-LIFE: 312.1 days<br><b>DECAY EMISSIONS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                    |                                                                                                                                                                                                                                                                                             |         |   |
| Gammas / X-rays                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     | Betas / Positrons (+) / Electrons* |                                                                                                                                                                                                                                                                                             | Alphas  |   |
| E (keV)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | %   | E (keV, Ave)                       | %                                                                                                                                                                                                                                                                                           | E (keV) | % |
| 835                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 100 |                                    |                                                                                                                                                                                                                                                                                             |         |   |
| - Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |                                    |                                                                                                                                                                                                                                                                                             |         |   |
| <b>STANFORD HAZARD CATEGORY</b><br>C – level (low hazard): ≤ 20 mCi<br>B – level (moderate hazard): > 20 mCi ≤ 1 Ci<br>A – level (high hazard): > 1 Ci                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |                                    |                                                                                                                                                                                                                                                                                             |         |   |
| <b>EXTERNAL RADIATION HAZARDS</b><br>Gamma dose rate, point source at 1 ft, 1 mCi:<br><b>4.4 mrem/h</b><br>Beta dose rate to skin, point source at 1 ft, 1 mCi:<br><b>0 mrem/h</b><br>Contamination skin dose, uniform deposit of 1 µCi per cm²:<br><b>230 mrem/h</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                    | <b>INTERNAL RADIATION HAZARDS</b><br>Annual Limit on Intake: <b>2000 µCi</b> (Ingestion)<br><b>800 µCi</b> (Inhalation)<br><br>The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE). |         |   |
| <b>SHIELDING</b><br><b>Gammas/X-rays:</b><br><b>3 cm</b> of lead will reduce the gamma dose rate by 90%.<br><br><b>Betas/electrons:</b><br>N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |                                    | <b>DOSIMETRY AND BIOASSAY REQS</b><br>Whole-body and finger-ring dosimeters are required for handling <b>5 mCi</b> or more, or <b>1 mCi amounts weekly</b> . Urine assays may be required after large spills or contaminations.                                                             |         |   |
| <b>SPECIAL PROBLEMS AND PRECAUTIONS:</b> <ol style="list-style-type: none"> <li>1. Recommended survey probe: <b>NaI</b></li> <li>2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.</li> <li>3. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required.</li> <li>4. Segregate waste to those with half-lives greater than <b>120 days (excluding H3 and C14)</b>. Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot.</li> <li>5. Limit soluble waste to the sewer to less than <b>100 µCi/day</b> per lab.</li> </ol> |     |                                    |                                                                                                                                                                                                                                                                                             |         |   |

**References:**

- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2<sup>nd</sup> ed.). Ashford, Kent: Nuclear Technology Publishing.
- Johnson, T.E., Birkby, B.K. (2012). Health Physics and Radiological Health (4<sup>th</sup> ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
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- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>